

Stochastic Gravitational-Wave Backgrounds

from a random metric to the Hellings-Downs curve

$$\Gamma_{ab} = \frac{3}{2} x_{ab} \ln x_{ab} - \frac{1}{4} x_{ab} + \frac{1}{2} + \frac{1}{2} \delta_{ab}$$



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STEP 1

Plane-wave expansion & pulsar response

$$h_{ij}(t, \vec{x}) = \sum_A \int_{-\infty}^{\infty} df \int_{S^2} d^2\Omega \tilde{h}_A(f, \hat{\Omega}) e^{2\pi i f(t - \hat{\Omega} \cdot \vec{x})} e_{ij}^A(\hat{\Omega})$$

$$\text{recall: } z(t, \hat{\Omega}) = \frac{1}{2} \frac{p^i p^j}{1 + \hat{\Omega} \cdot \hat{p}} \Delta h_{ij}$$

Each pulsar's response is an antenna pattern F^A acting on the wave.

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STEP 2

A stochastic signal: Gaussian random fields

Sum over the whole population $\Rightarrow$ treat $\tilde{h}_A$ as zero-mean Gaussian fields,
fully described by their second moments (the variance).

With Gaussian statistics, the background lives entirely in its power spectrum.



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$$\langle \tilde{h}_A(f, \hat{\Omega}) \tilde{h}_{A'}^*(f', \hat{\Omega}') \rangle = \delta(f - f') \delta^2(\hat{\Omega}, \hat{\Omega}') P_{AA'}(f, \hat{\Omega})$$

stationary

directions
uncorrelated

polarization
tensor

With Gaussian statistics, the background lives entirely in its power spectrum.

STEP 3

Polarization: the Stokes parameters

$$P_{AA'} = \begin{bmatrix} I+Q & U-iV \\ U+iV & I-Q \end{bmatrix}$$

$S_h(f)$: one-sided PSD. $P(\hat{\Omega})$: sky distribution ($=1$ if isotropic).

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$$\text{unpolarized} \Rightarrow Q=U=V=0 \Rightarrow P_{AA'} = I(f, \hat{\Omega}) \delta_{AA'}$$

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isotropic, one-sided PSD S_h :

$$\langle \tilde{h}_A \tilde{h}_{A'}^* \rangle = \frac{\delta_{AA'}}{2} \frac{\delta(f-f')}{2} \frac{\delta^2(\hat{\Omega}, \hat{\Omega}')}{4\pi} S_h(f) P(\hat{\Omega})$$

$S_h(f)$: one-sided PSD. $P(\hat{\Omega})$: sky distribution (=1 if isotropic).

STEP 4

Cross-correlate two pulsars (isotropy)

$$\langle \tilde{z}_a(f) \tilde{z}_b^*(f') \rangle = \frac{1}{2} \delta(f - f') S_z(f)_{ab}$$

S_z is the cross-power spectral density of the pulsar redshifts.

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$$K_{ab} = \left(\exp \left[-2\pi i f L_a (1 + \hat{\Omega} \cdot \hat{p}_a) \right] - 1 \right) \left(\exp \left[+2\pi i f L_b (1 + \hat{\Omega} \cdot \hat{p}_b) \right] - 1 \right)$$

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The pulsar term decorrelates

$$fL \gg 1 \quad (f \sim 10^{-9} \text{ Hz}, \quad L \sim 100 \text{ pc})$$

Only the Earth terms stay correlated between distinct pulsars.



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$$K_{ab}(f, \hat{\Omega}) \longrightarrow \begin{cases} 2 & (a = b), \\ 1 & (a \neq b) \end{cases}$$

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STEP 6

The overlap reduction function → Hellings–Downs

$$\tilde{\Gamma}_{ab} = \int_{S^2} \frac{d^2\hat{\Omega}}{4\pi} \sum_A F_a^A F_b^A = x_{ab} \ln x_{ab} - \frac{1}{6} x_{ab} + \frac{1}{3}$$

Quadrupolar correlation vs. angular separation — the smoking gun.



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renormalize so $\Gamma_{ab} = 1$ at $a = b$:

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$$x_{ab} = \frac{1}{2}(1 - \cos \theta_{ab}), \quad \Gamma(90^\circ) \approx -0.15, \quad \Gamma(180^\circ) = 0.25$$

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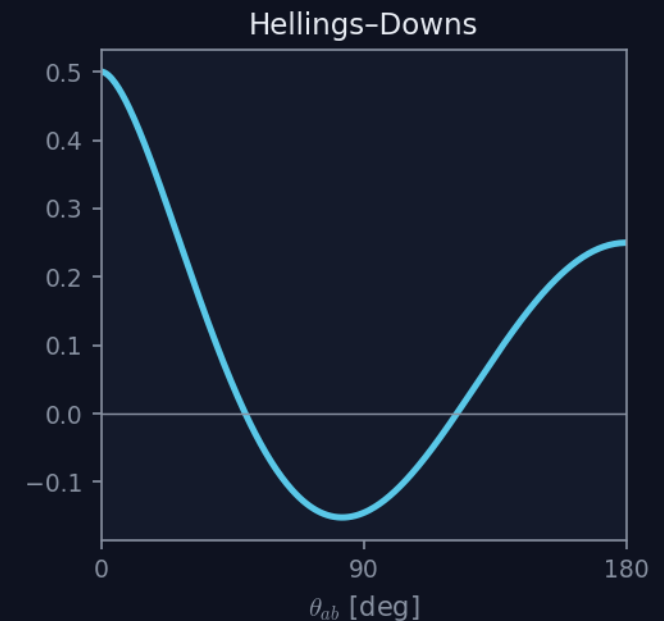
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STEP 7

From rates to residuals: the timing spectrum

$$R(t) = \int_0^t z(t') dt' \Rightarrow \text{factor } \frac{1}{2\pi i f}$$

recall $S_z(f)_{ab} = \frac{1}{3} \Gamma_{ab} S_h(f)$

Timing residuals turn the redshift spectrum into a steep $f^{-\gamma}$ law.

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$$S_t(f)_{ab} = \Gamma_{ab} \frac{A^2}{12\pi^2} f_{\text{yr}}^{-3} \left(\frac{f}{f_{\text{yr}}} \right)^{-\gamma}, \quad \gamma = 3 - 2\alpha$$

$$h_c(f) = A (f/f_{\text{yr}})^\alpha : \quad \text{SMBHBs } \alpha = -\frac{2}{3}, \gamma = \frac{13}{3} \quad \text{relic } \alpha = -1, \gamma = 5$$

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STEP 8

One binary: loudness $\times$ time in band

single binary (circular, GW-driven):

$$\frac{dE}{d\ln f_r} = \frac{dE}{dt} \cdot \frac{dt}{d\ln f_r}$$

The energy per log-frequency = per-binary loudness $\times$ time spent in band.



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$$\Rightarrow \frac{dE}{d\ln f_r} \propto (G\mathcal{M}_c)^{5/3} f_r^{2/3}$$

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STEP 9

The population sum: $h_c \propto f^{-2/3}$

cosmic population sum (Phinney 2001):

$$h_c^2(f) = \frac{4G}{\pi c^2 f^2} \int_0^\infty dz \frac{dN}{dz} \frac{1}{1+z} \frac{dE}{d \ln f_r} [f_r = f(1+z)]$$

Summing GW-driven binaries over cosmic time gives the $-2/3$ power law.

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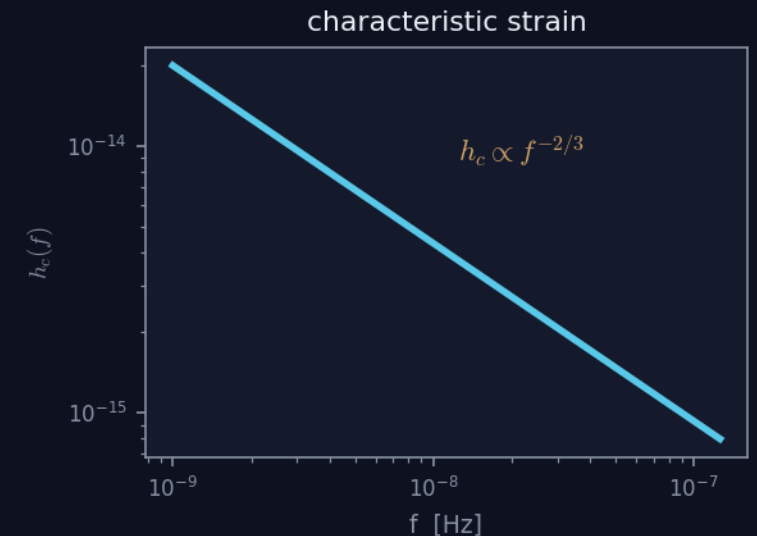
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$$h_c(f) = A(f/f_{\text{yr}})^{-2/3}$$



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Energy density & source populations

$$\Omega_{\text{GW}}(f) = \frac{1}{\rho_c} \frac{d\rho_{\text{GW}}}{d\ln f} = \frac{2\pi^2}{3H_0^2} f^2 h_c^2(f)$$

The spectral slope is a direct readout of the source population.

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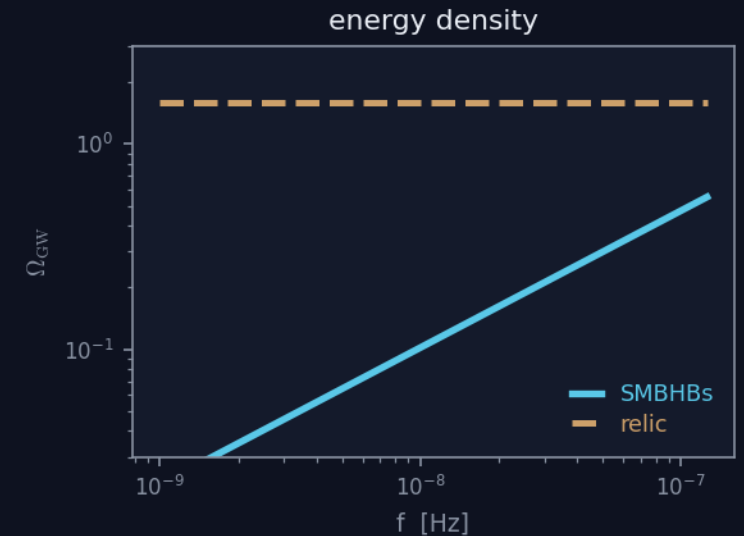
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